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Performance Balance

- While processor power has raced ahead at breakneck speed, other critical components of the computer have not kept up.
- The result is a need to look for performance balance: an adjusting of the organization and architecture to compensate for the mismatch among the capabilities of the various components.
- The problem created by such mismatches is particularly critical at the interface between processor and main memory.
- While processor speed has grown rapidly, the speed with which data can be transferred between main memory and the processor has lagged badly.
- The interface between processor and main memory is the most crucial pathway in the entire computer because it is responsible for carrying a constant flow of program instructions and data between mem chips and the processor.
- If memory or the pathway fails to keep pace with the processor's insistent demands, the proce in a wait state, and valuable processing time is lost.

Performance Balance

- A system architect can attack this problem in a number of ways, all of which are reflected in contemporary computer designs. Consider the following examples:
  - Increase the number of bits that are retrieved at one time by making DRAMs "wider" rather than "deeper" and by using wide bus data paths.
  - Change the DRAM interface to make it more efficient by including a cache or other buffering scheme on the DRAM chip.
  - Reduce the frequency of memory access by incorporating increasingly complex and efficient cache structures between the processor and main memory.
    - This includes the incorporation of one or more caches on the processor chip as well as on an off-chip cache close to the processor chip.
  - Increase the interconnect bandwidth between processors and memory by using higher-speed buses and a hierarchy of buses to buffer and structure data flow.

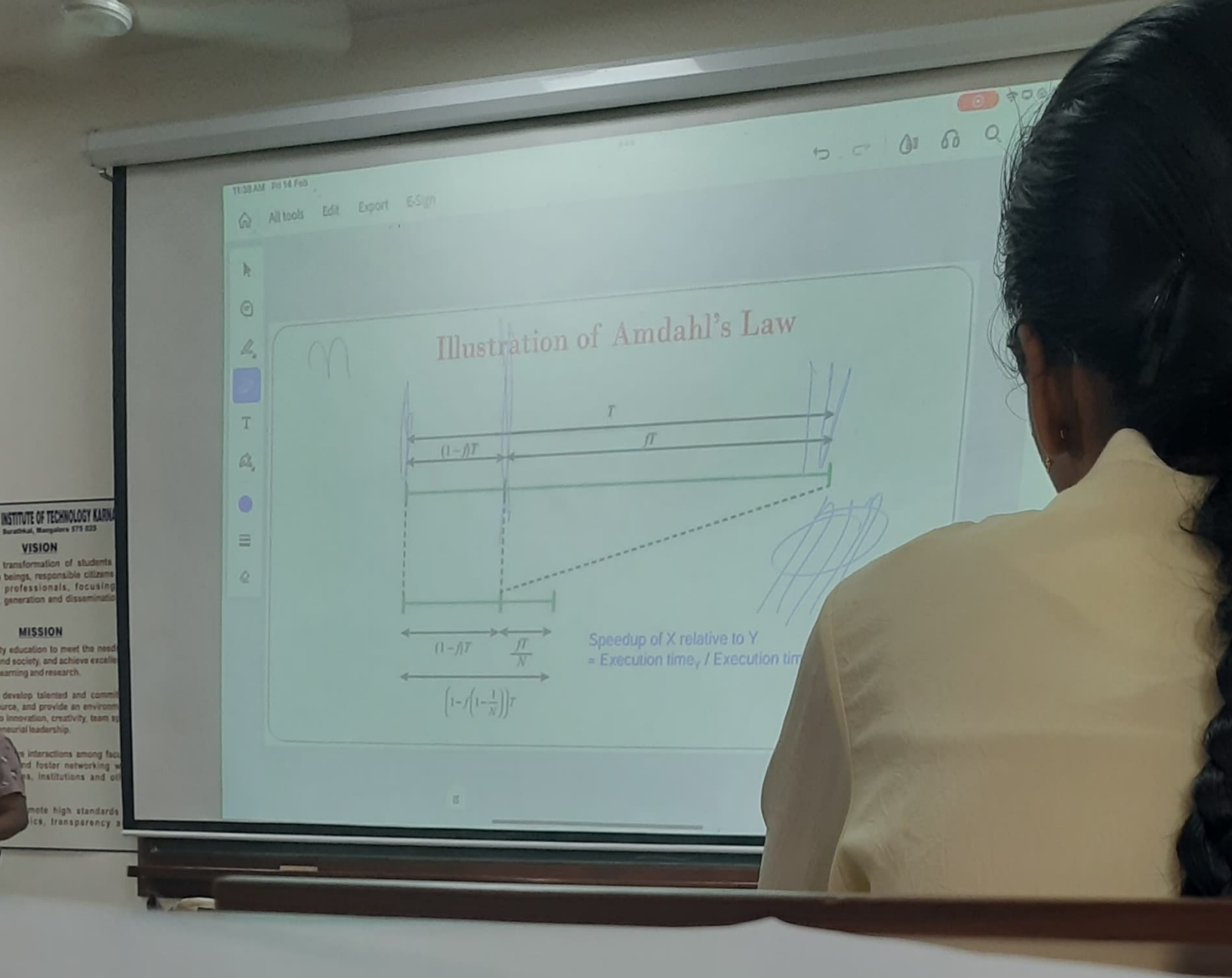
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Illustration of Amdahl's Law
T
(1 − f)T
fT
(1 − f)T
fT/N
(1 − f(1 − 1/N))T
Speedup of X relative to Y
= Execution time_Y / Execution tim
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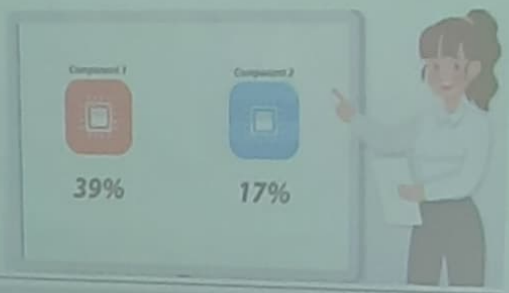
Amdahl's Law: Example

- Suppose that a task makes extensive use of floating-point operations, with 40% of the time consumed by floating-point operations. With a new hardware design, the floating-point module is sped up by a factor of K.
- Then the overall speedup is 1.67



Superscalar Execution

This is the ability to issue more than one instruction in every processor clock cycle. (In effect, multiple parallel pipelines are used.)

IF	ID	EX	MEM	WB				
IF	ID	EX	MEM	WB				
	IF	ID	EX	MEM	WB			
	IF	ID	EX	MEM	WB			
		IF	ID	EX	MEM	WB		
		IF	ID	EX	MEM	WB		
			IF	ID	EX	MEM	WB	
			IF	ID	EX	MEM	WB	
				IF	ID	EX	MEM	WB
				IF	ID	EX	MEM	WB

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Amdahl's Law

- *Computer system designers look for ways to improve system performance by advances in technology or change in design.*
- *Examples include the use of parallel processors, the use of a memory cache hierarchy, and speedup in memory access time and I/O transfer rate due to technology improvements.*
- *In all of these cases, it is important to note that a speedup in one aspect of the technology or design does not result in a corresponding improvement in performance.*
- This limitation is succinctly expressed by Amdahl's law.
- Amdahl's law was first proposed by Gene Amdahl in 1967.
- *The law deals with the potential speedup of a program using multiple processors compared to a single pro*
- Nevertheless, Amdahl's law illustrates the problems facing industry in the development of m machines with an ever-growing number of cores:
- The software that runs on such machines must be adapted to a highly parallel execution env exploit the power of parallel processing.

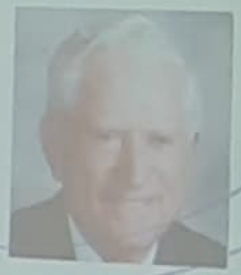
Amdahl's Law

- Consider a program running on a single processor such that a fraction (1 - f) of the execution time involves code that is inherently sequential, and a fraction f that involves code that is infinitely parallelizable with no scheduling overhead.
- Let T be the total execution time of the program using a single processor.
- Then the speedup using a parallel processor with N processors that fully exploits the parallel portion of the program is as follows:

$$\text{Speedup} = \frac{\text{Time to execute program on a single processor}}{\text{Time to execute program on } N \text{ parallel processors}}$$

$$= \frac{T(1-f) + Tf}{T(1-f) + \frac{Tf}{N}} = \frac{1}{(1-f) + \frac{f}{N}}$$

amdahl



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Amdahl's Law (Fixed Size Speedup)

- Let f be the fraction of a program that is sequential
  - $1-f$ is the fraction that can be parallelized
- Let T_1 be the execution time on 1 processor
- Let T_p be the execution time on p processors
- S_p is the *speedup*

$$
\begin{aligned}
S_p &= T_1 / T_p \\
&= T_1 / (fT_1 + (1-f)T_1 / p) \\
&= 1 / (f + (1-f)/p)
\end{aligned}
$$

- As $p \to \infty$

$$S_p = 1 / f$$

$f \to$

Comparing Computer Performance

- To compare the performance of two machines (or CPUs) "A", "B" running <u>a given specific program</u>:

$$\text{Performance}_A = 1 \;/\; \text{Execution Time}_A$$
$$\text{Performance}_B = 1 \;/\; \text{Execution Time}_B$$

- *Machine A is n times faster than machine B means* (or slower? if n < 1)

$$\boxed{\text{Speedup} = n = \frac{\text{Performance}_A}{\text{Performance}_B} = \frac{\text{Execution Time}_B}{\text{Execution Time}_A}}$$

Comparing Computer Performance: Example

- For a given program:

  Execution time on machine A: $\text{Execution}_A = 1$ second

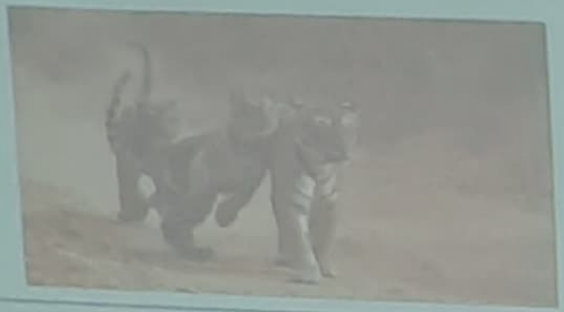
  Execution time on machine B: $\text{Execution}_B = 10$ seconds

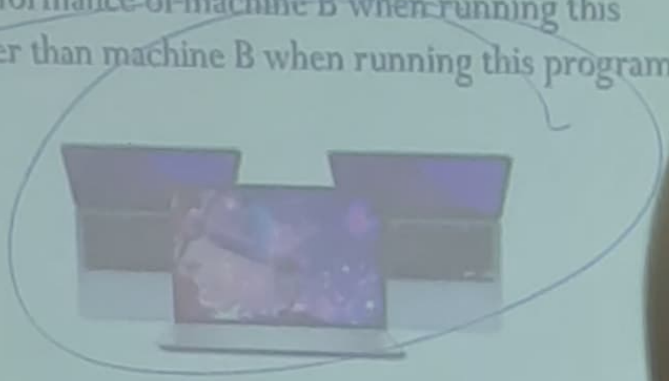
  Speedup = Performance_A / Performance_B

  = Execution Time_B / Execution Time_A

  = 10 / 1 = 10

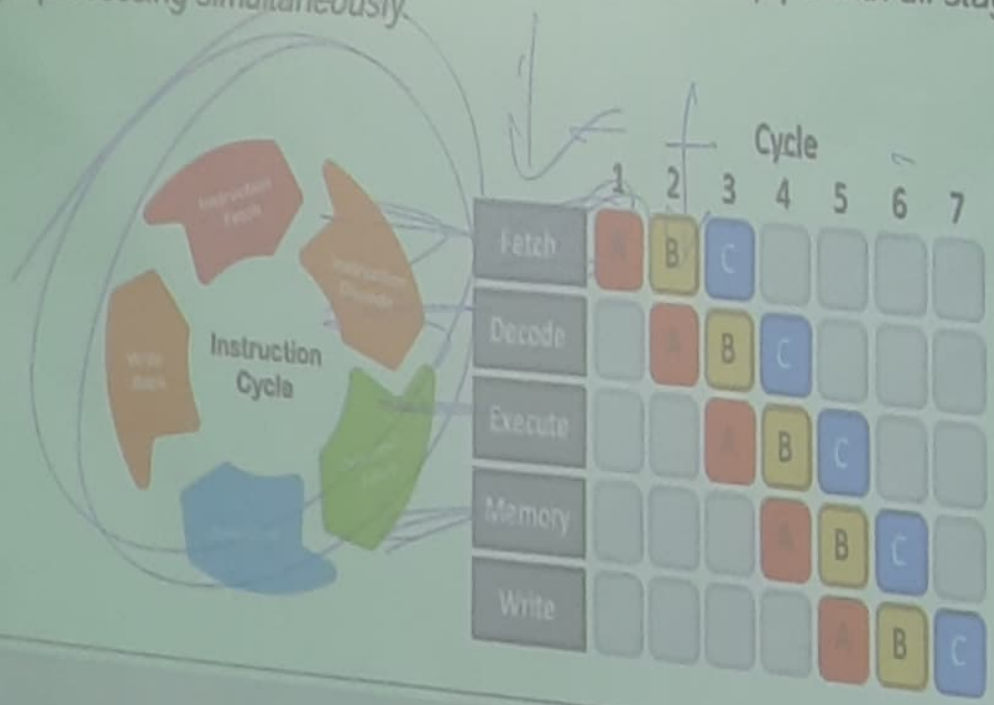
- The performance of machine A is 10 times the performance of machine B when running this program, or: Machine A is said to be 10 times faster than machine B when running this program.





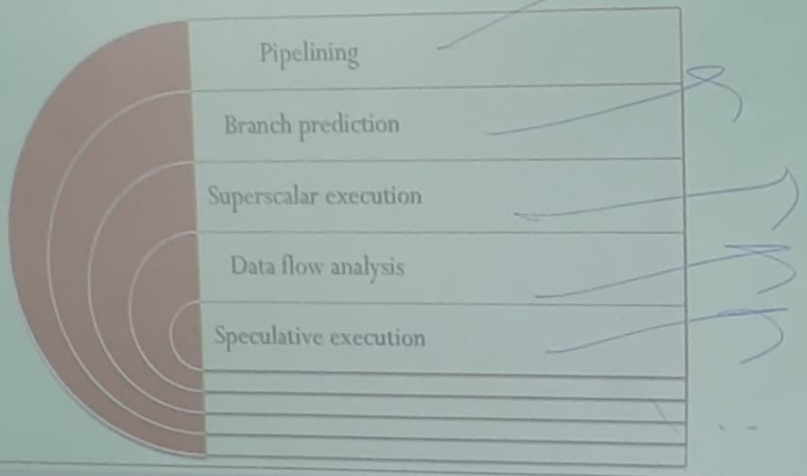
Pipelining

Processor moves data or instructions into a conceptual pipe with all stages of the pipe processing simultaneously.



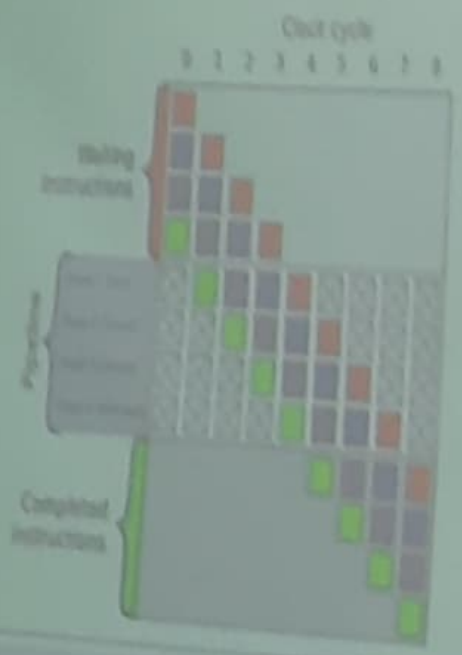
Microprocessor Speed

Techniques built into contemporary processors include:



Branch Prediction

Processor looks ahead in the instruction code fetched from memory and predicts which branches, or groups of instructions, are likely to be processed next.



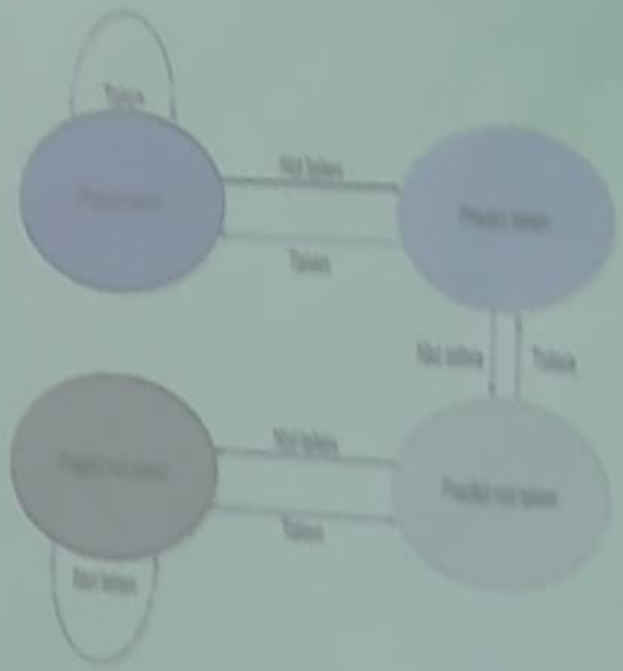


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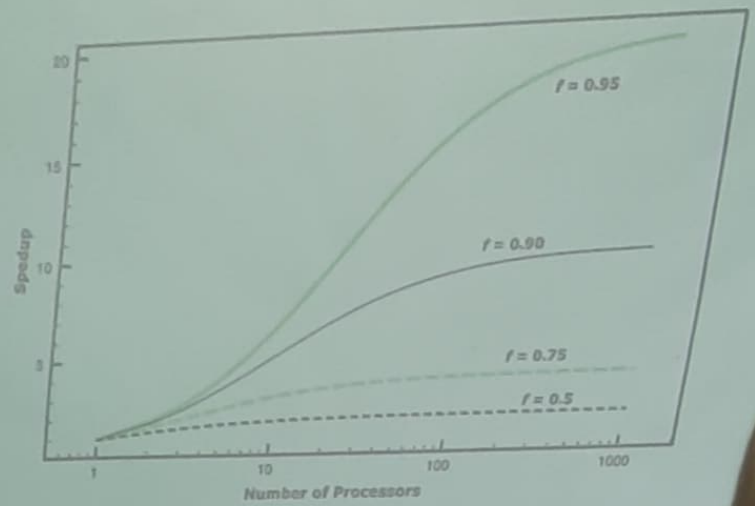
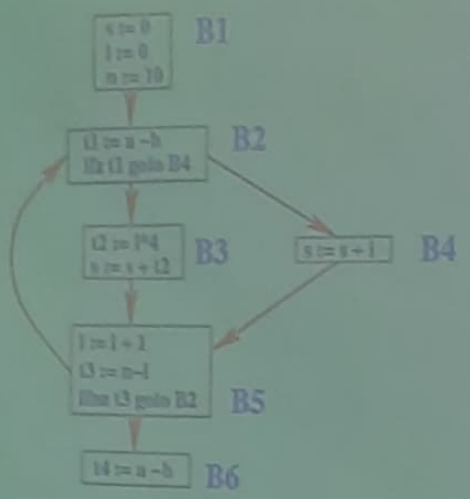
Illustration of Amdahl's Law



Data Flow Analysis



Processor analyzes which instructions are dependent on each other's results, or data, to create an optimized schedule of instructions.